THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **A Square and a Cube, Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 400	$20^2 = 400$
2	**(a)** 625	$25 \times 25 = 625$
3	**(a)** 23	$23 \times 23 = 529$
4	**(b)** 64	Sum of first n odd numbers $= n^2 = 64$
5	**(d)** 14	There are $2n = 14$ numbers between n^2 and $(n+1)^2$
6	**(a)** 5^9	Add the exponents: $2 + 7 = 9$
7	**(d)** 3^4	Subtract the exponents
8	**(a)** 2^{12}	Multiply the exponents
9	**(c)** $\frac{1}{8}$	$a^{-n} = \frac{1}{a^n}$
10	**(d)** $\frac{1}{16}$	$a^{-n} = \frac{1}{a^n}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A perfect square cannot end in 2, 3, 7 or 8 (1); 2468 ends in 8 (1).	1+1
12	$2744 = 2^3 \times 7^3$ (1); cube root $= 2 \times 7 = 14$ (1).	1+1
13	$a^n \div a^n = 1$ and by the law also $a^{n-n} = a^0$ (2).	2
14	$2^{10} = 1024 > 100 = 10^2$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Side $= \sqrt{1296} = 36$ m (2); perimeter $= 4 \times 36 = 144$ m (1).	2+1
16	3.27×10^{-5} (2); four zeros (1).	2+1
17	Volume $= 13.824$ m³ (1); edge $= \sqrt[3]{13.824} = 2.4$ m (2).	1+2
	OR: $T_n + T_{n+1} = (n+1)^2$ (1); $10 + 15 = 25 = 5^2$ (1); also $6 + 10 = 16$ (1).	1+1+1
18	1.5×10^8 km (2); 15 crore km (1).	2+1
	OR: $\left(\frac{4}{3}\right)^2$ (1.5); $\frac{16}{9}$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $10^6 - 10000 + 25 = 990025$ (2.5). (ii) $10000 - 600 + 9 = 9409$ (2.5).	2½+2½
	OR: (i) Ones digit restrictions; square of even is even and of odd is odd; $n^2 - (n-1)^2 = 2n - 1$ (3). (ii) Examples such as $12^2 = 144$, $13^2 = 169$, $9^2 - 8^2 = 17$ (2).	3+2
20	(i) $a^{m-m} = a^0 = 1$ (3). (ii) $9 \times \frac{1}{9} = 1$ (2).	3+2
	OR: (i) $2^5 = 32$ (2). (ii) $(-2)^{3-5+1} = (-2)^{-1} = -\frac{1}{2}$ (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapters 1 & 2: **A Square and a Cube, Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 729	$9 \times 9 \times 9 = 729$
2	**(b)** 9	$9^3 = 729$
3	**(b)** 125	$5^3 = 125$
4	**(c)** smallest number expressible as the sum of two cubes in two ways	$1729 = 1^3 + 12^3 = 9^3 + 10^3$
5	**(b)** 2	$72 = 2^3 \times 3^2$; one more 2 is needed
6	**(c)** $\frac{8}{125}$	$\frac{2^3}{5^3}$
7	**(d)** 6.3×10^6	One non-zero digit before the decimal point
8	**(a)** 10^7	1 crore = 100 lakh = 10,000,000
9	**(d)** 75000	Move the decimal 4 places right
10	**(c)** 6	$64 = 2^6$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Three natural numbers a, b, c with $a^2 + b^2 = c^2$ (1); for example $6, 8, 10$ (1).	1+1
12	It is the smallest number that is the sum of two cubes in two ways: $1^3 + 12^3 = 9^3 + 10^3$ (2).	2
13	3.5×10^6 (2).	2
14	3×10^8 m/s (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The digits rise to the number of ones, then fall back (1); $1111^2 = 1234321$ (1); $11111^2 = 123454321$ (1).	1+1+1
16	$\frac{1}{8} + \frac{1}{9}$ (1.5); $\frac{17}{72}$ (1.5).	1½+1½
17	$(80 - 2)^2 = 6400 - 320 + 4$ (2); $= 6084$ (1).	2+1
	OR: $1372 = 2^2 \times 7^3$ (1); not a cube since 2^2 is not a triple (1); multiply by 2 (1).	1+1+1
18	$2^6 \times 2^{-4} = 2^2 = 4$ (3).	3
	OR: Earth is greater (1); ratio ≈ 81 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let sides $3x, 2x$: $6x^2 = 1176$ (2); $x^2 = 196$, $x = 14$ (2); sides 42 m and 28 m (1). Perimeter 140 m, so fencing costs ₹2800.	2+2+1
	OR: (i) $6 \times 7 = 42 = \sqrt{1764}$ (2). (ii) $2916 = 2^2 \times 3^6$, root $= 2 \times 27 = 54$ (3).	2+3
20	(i) $10^{-5}/10^{-10} = 10^5$ times (3). (ii) 7×10^{-3} mm (2).	3+2
	OR: (i) Compare the powers of 10 first, then the decimal parts (3). (ii) $4.5 \times 10^7 = 45 \times 10^6 > 9.1 \times 10^6$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **A Square and a Cube, Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 81	$9^2 = 81$
2	**(c)** 1296	$36 \times 36 = 1296$
3	**(a)** 26	$26 \times 26 = 676$
4	**(d)** 144	Sum of first n odd numbers $= n^2 = 144$
5	**(d)** 22	There are $2n = 22$ numbers between n^2 and $(n+1)^2$
6	**(a)** 7^6	Add the exponents: $3 + 3 = 6$
7	**(d)** 5^5	Subtract the exponents
8	**(d)** 5^6	Multiply the exponents
9	**(c)** $\frac{1}{9}$	$a^{-n} = \frac{1}{a^n}$
10	**(a)** $\frac{1}{32}$	$a^{-n} = \frac{1}{a^n}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2 \times 15 = 30$ numbers lie between them (2).	2
12	$1521 = 3^2 \times 13^2$ (1); so $\sqrt{1521} = 3 \times 13 = 39$ (1).	1+1
13	$10^{5-2-2} = 10^1 = 10$ (2).	2
14	$\frac{1}{2^6} = 2^{-6}$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM $= 24 = 2^3 \times 3$ (1); need $2^4 \times 3^2$ (1); least square $= 144$ (1).	1+1+1
16	$2^{10} = 1024$ (3).	3
17	$1620 = 2^2 \times 3^4 \times 5$ (2); 5 has no pair, so it is not a perfect square (1).	2+1
	OR: 1, 8, 27, 64, 125 (2); odd cubes are 1, 27, 125 (1).	2+1
18	LHS $= \frac{1}{10}$; RHS $= \frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$ (3).	3
	OR: 2×10^3 (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{1225} = 35$ (1). (ii) 1296 is 36^2, so yes with 36 rows (2). (iii) $36^2 - 35^2 = 71$ (2).	1+2+2
	OR: (i) $14^2 = 196$, $15^2 = 225$, so between 14 and 15 (2). (ii) $14.1^2 = 198.81$; $14.2^2 = 201.64$; so about 14.1 (3).	2+3
20	(i) $\frac{2^8 \times 2^3}{2^6} = 2^5 = 32$ (3). (ii) 2^5 (2).	3+2
	OR: (i) 4.8×10^7 (1). (ii) $4.8 \times 10^7 + 5 \times 10^6 = 5.3 \times 10^7$ (2). (iii) 5.3×10^7 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1331	$11 \times 11 \times 11 = 1331$
2	**(a)** 12	$12^3 = 1728$
3	**(c)** 512	$8^3 = 512$
4	**(c)** even	even × even = even
5	**(a)** 4	$54 = 2 \times 3^3$; 2×2 more
6	**(d)** 4.5×10^3	One non-zero digit before the decimal point
7	**(a)** 8×10^{-5}	One non-zero digit before the decimal point
8	**(c)** 10^5	1 lakh = 100,000
9	**(c)** 0.002	Move the decimal 3 places left
10	**(d)** 1	Anything non-zero to power 0

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	There are $2n$ such numbers (2).	2
12	$105^2 = (10 \times 11) \mid 25 = 11025$ (2).	2
13	$625 = 5^4$ so $x = 4$ (2).	2
14	$30000 < 200000$, so 2×10^5 is larger (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$12^2 = 144$, $12^3 = 1728$ (2); difference 1584 (1).	2+1
16	$81 = 3^4$ so $x + 1 = 4$, $x = 3$ (2); $3^4 = 81$ (1).	2+1
17	Factors pair up except the square root (1.5); 36 has 9 factors: 1, 2, 3, 4, 6, 9, 12, 18, 36 (1.5).	1½+1½
	OR: LCM $= 60 = 2^2 \times 3 \times 5$ (1); need $2^2 \times 3^2 \times 5^2$ (1); square $= 900$ (1).	1+1+1
18	$5 \times 10^{-4} < 3 \times 10^{-3} < 2 \times 10^{-2}$ (3).	3
	OR: $0.1 \times 10^4 = 1000$ mm $= 1$ m (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) -12 (1.5). (ii) 0.3 and $\frac{3}{5}$ (3.5).	1½+3½
	OR: (i) $169 - 25 = 144 = 4 \times 36$ (3). (ii) $(13 + 5)(13 - 5) = 144$ (2).	3+2
20	(i) $6^3 = 216 = 8 \times 27$ (2). (ii) $(12/6)^3 = 8$ (3).	2+3
	OR: (i) Compact, easy to compare and calculate (2). (ii) 9.1×10^{-31} kg (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.